

OPTOMETRY

Perspectives on corneal topography: a review of videokeratoscopy

Peter R Keller BAppSc(Optom)

Paul P van Saarloos PhD

Centre for Ophthalmology and Visual Science, Lions Eye Institute

This article reviews the optical principles of computer-assisted videokeratoscopy (CVK) and provides a guide to the differences between short and long working distance systems. The description of the corneal surface involves a number of complex issues which have yet to be adequately resolved and the importance of alignment and reference axis assumptions to CVK is discussed. With the increasing clinical use of such systems, the debate of such issues has meaning, not just in the research environment but also in routine clinical practice. A number of applications are illustrated.

(*Clin Exp Optom* 1997; 80: 1: 18-30)

Accepted for publication: 27 January 1997

Key words: alignment, computer-assisted videokeratoscopy, Placido disc

The purpose of this article is to review the optical principles of computer-assisted videokeratoscopy and to comment briefly on the clinical applications of such instruments. The demand for precise clinical corneal topography data has shifted from the diagnosis of corneal disease, through contact lens management, to the new forms of keratorefractive surgery. Although each application sets slightly different information demands, an understanding of the important assumptions and limitations of the current technology may lead to improved clinical interpretation of the output of corneal topography system.

THE CORNEA

The cornea plays a central role in both the structural integrity and the refractive performance of the eye.¹ Based on various schematic eyes,² the anterior corneal surface contributes more than two-thirds

of the optical power of the entire human eye. Through accurate measurement of the topography of the anterior corneal surface it is possible to predict the contribution of this surface to retinal image formation.³⁻⁵ This applies to simple spherocylindrical refractive powers as well as primary and higher-order aberrations.

The refractive role of the cornea is defined by its clarity and topographic characteristics, namely, surface curvature and regularity. Measurement of corneal shape parameters has evolved from qualitative keratoscopy, through keratometry and photokeratoscopy, to computer-assisted corneal topography (CT). Corneal topography can be measured by a number of different techniques using either the virtual or real image of a target projected onto the cornea. Of these, videokeratoscope (VK) based systems have demonstrated clinical popularity for the measurement of corneal shape and the detection of subtle changes in corneal contour.⁶⁻¹³

This emerging technology is not without practical limitations and controversy. Much of the confusion arises from the inconsistent and incorrect use of terminology.^{9,14,15} In any discussion of the corneal topography literature, it is important to clearly establish a standard set of definitions to avoid confusion.

ALIGNMENT AND CENTRATION

Topography systems provide detailed information about the corneal surface from which visual performance characteristics can be calculated (see below). Central to instrument design, in both hardware and software, are several key assumptions. The most important of these assumptions is the location of a reference axis. Reference axes are commonly encountered in geometrical optics, in which case the meanings are, with few exceptions, unequivocal.¹⁶ However, because the human eye is not a centred optical system, a number of

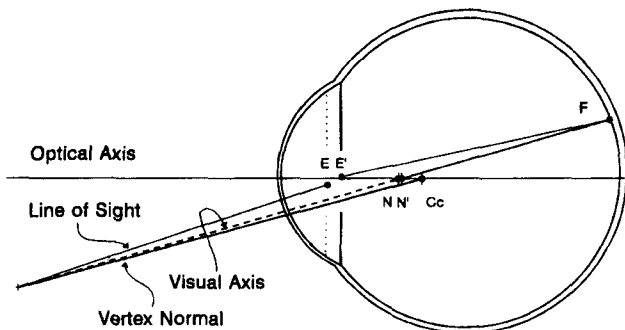


Figure 1. Schematic of the important eye reference points and axes. Entrance pupil centre (E), exit pupil centre (E'), fovea (F), nodal points (N and N') and a corneal centre of curvature (Cc)

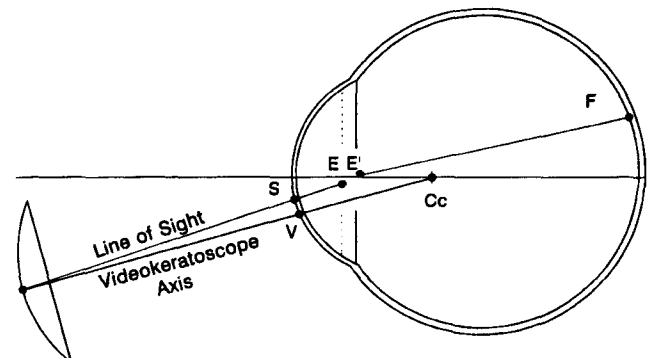


Figure 2. Standard videokeratoscope alignment with the instrument coaxial with the vertex (V) normal. The line of sight typically does not strike the cornea at this same point but slightly temporal to it at the point S, sometimes called the corneal sighting centre

theoretical and experimental axes exist which can be used to relate eye alignment and visual performance and it is here that potential confusion lies. Although schematic eyes are useful theoretical constructs they do not reveal the dynamic interactions between the visual reference axis, pupil centration, retinal directional sensitivity and monochromatic aberrations. The effect of these reference axis assumptions on CT power maps has been debated¹⁷⁻²⁴ but consensus is yet to be achieved.

The dichotomy of alignment and centration is of particular significance, not only in the measurement of corneal topography but also in centration strategies for keratorefractive procedures and the modelling of visual performance. It is within this context that the following terms are defined.

VISUAL REFERENCE AXIS

The selection of a suitable visual reference axis may seem elementary. The historical alternatives are the line of sight and the visual axis but even these two terms possess a number of conflicting definitions. The selection of a visual reference axis for corneal topography is important through-

out the CT process, in understanding instrument alignment and positioning, in the assumptions inherent to the surface reconstruction algorithms and the manner in which the results are represented, particularly when deriving visual performance measures.

The visual axis is useful in determining image magnification in Gaussian optics. However, it is not the most appropriate axis for either measuring or reporting corneal topography. When describing the corneal contribution to retinal image formation, the appropriate visual reference axis would describe the actual path light follows from the fixation target to the corneal surface and then, by refraction, through the eye to strike the fovea. This would avoid resorting to schematic eye models and could be measured both experimentally and clinically. The line of sight is thought to best fill this requirement, but videokeratoscope systems do not measure corneal topography relative to this axis. The questions arise of how the various alternative reference axes differ and which is the most appropriate for measuring corneal topography.

The line from the fixation point to the centre of the entrance pupil (E) and from the exit pupil centre (E') to the

fovea (F) is the generally accepted definition of the line of sight^{11,25} (Figure 1). It has been suggested that it represents the central or chief ray of the bundle of light passing from the fixation point through the actual optics of the eye to the fovea.^{11,25-26} The entrance pupil is the virtual image of the real pupil, and a light ray directed at the entrance pupil is refracted by the cornea to pass through the real pupil.

The visual axis is defined as the line from the fixation point through the nodal points (N, N') to the fovea.²⁷ As it is defined here, it is a theoretical construct and its corneal intercept cannot be found using routine clinical testing methods. Mandell²⁶ and others^{28,29} have suggested that a very close experimental approximation of the visual axis (also called the nodal axis²) is the achromatic axis, which is experimentally defined as the axis along which two split field targets illuminated in different wavelengths in dual Maxwellian view appear aligned.²⁸ In fact, a number of research papers have either reported the displacement of excimer ablations relative to the corneal intercept of the visual axis when in fact measuring it from the vertex^{9,14} or have claimed the two points to be the same.³⁰

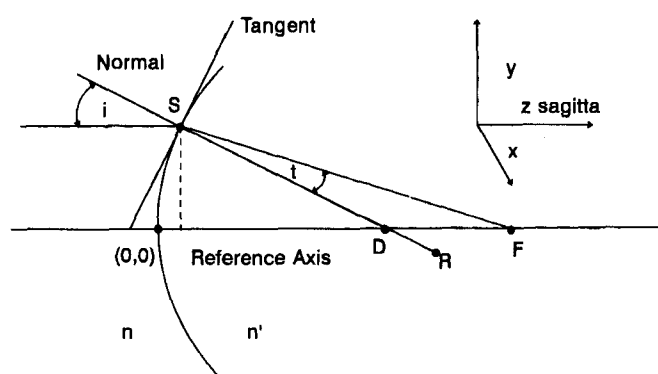


Figure 3. Corneal surface geometry of a meridional section. Surface point (S), centre of curvature (R), D is the axial intercept of normal to tangent at S, refractive indices (n and n'), angle of incidence (i) and reflection (t), and the secondary focal point (F)

The corneal light reflex is another potential alignment identification point and is readily observed both clinically and experimentally, although its position and significance are similarly misunderstood.^{9,14,30,31} Also known as the first Purkinje image, it is the virtual image formed by the anterior corneal surface acting as a partial mirror. Maloney¹¹ defined the corneal vertex as the point on the cornea closest to the fixation target, which is readily identified overlying the corneal light reflex. The vertex normal connects the fixation light, the corneal vertex, the light reflex and the corneal vertex centre of curvature. This definition differentiates the corneal vertex, which is directionally sensitive, from the apex which has been described as the region of greatest curvature. Although this terminology has not been universally accepted, the terms vertex and apex can otherwise be confusing, particularly when the region of greatest curvature is located in the mid-periphery after keratorefractive surgery. Mandell³² suggests that the term 'VK axis point' replace vertex, effectively using the same argument. However, this suggestion unnecessarily limits the more general non-videokeratoscope use of this terminology.

System alignment and centration in image capture

If videokeratoscope systems are aligned along an axis which is neither the visual axis nor the line of sight, then how are they aligned? The answer is surprisingly simple. The subject looks at a fixation light on the optical axis of the instrument and the VK is manipulated to achieve perpendicular alignment with the cornea. At this point the instrument is aligned with the corneal vertex normal and is thus directed towards that regional centre of corneal curvature (Cc in Figure 2). When the VK is correctly aligned the Placido rings are then reflected back around the instrument axis and captured by the video camera, producing a concentric ring image centred around the corneal vertex. The proprietary algorithms which reconstruct the corneal surface from the video image assume that the instrument axis and vertex normal are coincident, although various 'corrections' are possible. Realigning the VK so that the instrument axis is centred on the corneal intercept of the line of sight would result in the VK no longer being perpendicular to the corneal surface and the algorithms invalidated.²⁷

System alignment and centration in power map generation

A topographic colour-coded map is the current standard form of CT system output which provides a wealth of information about the corneal surface in a single display. Isodioptric maps are constructed using topographic data derived from the reconstructed corneal surface coordinates, which are calculated from the video image with *a priori* knowledge of the various instrument constants. The algorithms central to these calculations assume that the instrument is aligned with a normal to the corneal surface, otherwise they are invalidated and errors introduced.^{11,27}

A number of different radii and power values can be calculated from the same surface co-ordinates.^{23,24,27} Regardless of the relative merits of the various power alternatives, the default common to all the current commercial systems has been termed 'axial' power. Axial power is typically calculated using the keratometric formula and the distance from the surface point along its normal to the system reference axis (SD in Figure 3). Substitution of the instantaneous (also sometimes called 'local' or ambiguously 'tangential') radius of curvature (SR in Figure 3) at that surface point into the keratometric formula gives the instantaneous power. Roberts¹⁷⁻¹⁹ and Cohen and colleagues²² have criticised the spherical bias of CT algorithms. Although the clinical utility of referencing to a fixed rather than floating axis system has been described,^{20,21} Roberts' argument makes clear the importance of the assumptions regarding a reference axis in corneal topography system outputs.

Hubbe and Foulks¹⁰ measured the effect on corneal topography of aspheric test surface misalignment and eccentric subject fixation, concluding that improper alignment induced error in computer-assisted topographic analysis. Although their report contained an incorrect interpretation of the visual axis and its experimental measurement, their conclusion that misalignment of CT systems may mislead interpretation of power maps remains valid. Eccentric subject fixation

produced irregular astigmatism similar to that seen in keratoconus, and although the angles used were larger than those between the different reference axes, the importance of reference axis effects was demonstrated.

From the literature, it can be summarised that consensus has not yet been achieved on the effect of reference axis assumptions on corneal topography. It is difficult to argue against the adoption of a standard set of terms to clarify the core concepts, although it is recognised that it would be difficult to define and enforce such terminology. The clinical implications of reference axis effects generally are not well understood or appreciated but the issues have become clearer over time. Reconstruction of corneal topography outputs referenced to the line of sight may well be the answer, and the continuing development of hardware and software systems will enable the definitive investigation of the significance of such complex issues.

HISTORY OF CORNEAL CONTOUR MEASUREMENT

Interest in the corneal surface shape traces back several centuries but the principles of keratoscopy have not changed significantly since the 19th century work of Goode and Placido. These principles dictate that an appreciation of the corneal topography can be gathered by inspection of the image formed by the cornea acting as a partial mirror. Historically, an alternating black and white ring target, or Placido disc, has been the most common form of target. The technological advances in keratorefractive surgery and improvements in computer capabilities in the 1980s facilitated the latest improvements in the precise measurement of corneal topography. A similar interest surge in corneal topography measurement was prompted by the commercial development of contact lenses in the 1960s. Although a number of different techniques is available, including, rasterstereography, interference patterns (moiré fringes), holography, ultrasonography and videokeratoscope systems, it is this last group

of instruments that has enjoyed commercial prominence. Though they may seem different, the majority of the current range of videokeratoscope systems employ algorithms based on Gullstrand's equations³⁸ and/or the principles of keratoscopy.

The development of the keratometer has been reviewed by Clark³⁴ and, although there is some disagreement, Helmholtz is generally credited with its invention in 1854. The keratometer uses optical doubling techniques to precisely measure radii of curvature of a small paracentral area. There have been attempts to modify the keratometer (for example, small mire keratometer³⁵) to enable measurement of the peripheral corneal contour, but mostly these have been unsuccessful due to problems with localisation of measurement area and repeatability. The limitations of keratometry are well documented.^{27,34-39} Depending on the shape of the cornea measured, the keratometric radius of curvature is determined using the assumed spherical arc averaged across two points separated along a chord of between 2.0 mm and 3.5 mm. Only two points on each arc are sampled and the cornea is presumed to be spherical between these points which does not hold true for irregular (for example, keratoconus) or aspheric surface shapes.

Keratoscopy provides a qualitative assessment of the catoptric image formed by the cornea of the keratoscope faceplate and relative differences in corneal curvature are observed as variations in the spacing and regularity of the image rings. Both Placido and Gullstrand are credited with producing early keratoscope photographs while the latter's equations form the basis of quantitative keratoscopy. Because there is no unique solution, the equations have to be solved iteratively, a laborious error-prone task prior to recent improvements in computers, and accordingly, analysis was performed qualitatively. Videokeratoscopy simply substitutes the photographic camera with a video camera and rather than qualitatively assessing the image pattern, image analysis software allows quantitative determination of the ring

shapes and hence the corneal contour.

Of the other techniques for measuring corneal topography, rasterstereography has attracted the greatest attention in clinical settings. This technique uses a projected grid to illuminate the corneal surface and the pattern of the grid is determined by the topography. To enable the grid pattern to be detected on the corneal surface, fluorescein is instilled into the tear fluid layer and a cobalt blue light source used. The results from an early rasterstereographic corneal topographic system^{31,40} show great promise but at the time of this review, cannot theoretically match the precision of keratoscopic methods for regular corneal shapes.

Because videokeratoscope systems rely on the reflective properties of the corneal surface, all such systems encounter difficulties, when the virtual image is broken up due to irregular corneal shapes, tear fluid layer discontinuities, certain corneal disease or epithelial disorders. They also require patient co-operation by steady fixation, which limits their use in animal studies. However, as videokeratoscopes and other corneal topography systems continue to evolve and improve, many of the current technical limitations will be overcome.

FACTORS AFFECTING CORNEAL TOPOGRAPHY

A number of variables have been identified as potential influences on corneal topography. Clark⁴¹ reviewed the limitations of early corneal topography systems in his development of an autocollimating photokeratoscope and surveyed a number of factors thought capable of causing a measurable change in corneal topography,⁴² concluding that with the exception of time and mechanical pressure, the alleged effects due to such variables as accommodation, convergence and certain drugs were of doubtful validity. It should be noted that with few exceptions the studies, which Clark⁴¹ reviewed, were limited to measures of central corneal radii and very little information was reported, or available, regarding the periphery.

The normal cornea exhibits surprising topographic stability despite a range of

external and internal forces. The effects of convergence and accommodation on corneal topography have been shown to be negligible over small sample sizes but with significant individual variations.⁴³⁻⁴⁵ Although it has been suggested^{44,45} that the horizontal meridian flattens while the vertical meridian steepens in curvature with convergence, the group average changes were generally less than 0.25 D. Bannon⁴⁶ reported similar findings of a trend towards a decrease in horizontal power (flattening of curvature) of between 0.25 to 0.50 D on convergence but cautioned on the size and significance of the results. In his summary of dynamic astigmatism, Brzezinski⁴⁷ suggested that this result was counter-intuitive to the expected steepening of the horizontal corneal meridian on medial rectus contraction and lateral rectus relaxation during convergence. He offered Fairmaid's explanation⁴⁵ that the vertical recti and possibly the obliques may exert a greater deforming force on the vertical meridian of the cornea.

As an indication of the stability of corneal curvature over time, a number of studies have measured corneal astigmatism for different age groups. It is generally accepted⁴⁸⁻⁵⁰ that corneal astigmatism is predominantly with-the-rule, although a small shift towards against-the-rule develops over time.⁵¹ Both the influence of the upper eyelid⁵²⁻⁵³ and extra-ocular muscles⁵⁴ have been suggested as the origin of this change but neither mechanism adequately explains the large individual variations observed. The average Asian corneal curvature is steeper and exhibits larger change with time than the average Caucasian corneal curvature. It has been suggested that this is due to tightness of the superior lid and smaller palpebral fissure size,⁴⁸ but it has not been suggested that the relationship between refractive astigmatism and corneal astigmatism is different for Asian eyes.

More recent research seems to suggest that the cornea is resistant to short duration forces such as normal and forced blinks and applanation tonometry.⁵⁵ Sustained lid pressure has been shown to cause monocular diplopia due to distorted

corneal topography but this too was observed to decay over a relatively short recovery time.⁵⁶ It remains to be seen whether these effects will be confirmed by similar studies performed with computer-assisted corneal topography systems.

Even though the cornea demonstrates remarkable stability under normal circumstances, it can be significantly altered by a number of surgical and non-surgical techniques. Such procedures and techniques typically produce permanent changes to the corneal topography whether by design or misadventure and have driven the development of new and improved measurement systems. Post-operative corneal topography has become a standard reporting tool in keratorefractive surgery.^{6-8,10-13,15,57-61} In addition, it is used in a range of applications including the analysis of early changes in corneal astigmatism after both cataract surgery^{62,63} and trabeculectomy,⁶⁴ in the assessment of pterygia⁶⁵ and in corneal disease, particularly keratoconus.⁶⁶⁻⁷⁰ Without the ability to measure the corneal surface shape, subtle irregularities, such as central islands after excimer photorefractive keratectomy⁷¹ and sub-clinical keratoconus, could not be as comprehensively assessed.

It has been recognised that significant changes in corneal topography can be associated with contact lens wear. A sample of the multitude of research papers published prior to 1975 covering this topic was listed by Clark.⁴² The emphasis since that time has been the investigation of contact lens induced corneal warpage.⁷²⁻⁷⁴ Wilson and colleagues⁷² differentiated between the changes due to corneal oedema and those due to corneal warpage. A further distinction can be made between inadvertent warpage and planned corneal moulding such as in orthokeratology, both of which produce distinct topographic changes to the cornea.

MATHEMATICAL MODELS AND BASIC PRINCIPLES

The human cornea is not a simple spherical surface. It has been described mathematically as a prolate (flattening) ellipse^{27,75} and is commonly astigmatic. Two

distinct classes of simple geometrical surfaces, torics and ellipses, can be combined to model the anterior corneal surface. Of the different conic sections, elliptical curves provide the most commonly employed general model for the various corneal hemi-meridia because of greater freedom of fit, although the fit deteriorates in the periphery and is dependent on several assumptions. Even more complex models are capable of incorporating toricity with progressive curvature changes towards the periphery.⁷⁶ A convenient equation often encountered in corneal modelling and which generates conic sections is Baker's equation:⁷⁷

$$y^2 = 2 r_0 z - p z^2 \quad (1)$$

where r_0 = apical radius,⁷⁷ $p = 1 - e^2$, e = eccentricity (e and p are indices of the rate of flattening), y is the ordinate axis and z the abscissa and axis of revolution.

The average corneal apical radii are 7.87 ± 0.25 mm in the flat meridian and 7.70 ± 0.27 mm in the steep meridian with an average p of 0.83 ± 0.13 in the flat and 0.81 ± 0.16 in the steep meridian.⁷⁸ It has been found that the rate of flattening is not related to apical radius. The average corneal chord (often incorrectly termed diameter) is marginally larger horizontally, 12.89 mm,⁷⁹ than vertically. When specifying a general model, the cornea flattens towards the periphery, although the amount of variation from a sphere is relatively small through the central optical area and the rate of flattening varies greatly between individuals.

Another useful mathematical concept in corneal topography is the relationship between chord length and arc radius, called the sagitta (Figure 4). Given an arc A-C with chord AC, the height from the vertex of the arc to the centre of the chord is called the sagitta. The relationship in the case of a spherical arc is described by the formula:

$$r = y^2 / 2s + s / 2 \quad (2)$$

which can be rewritten to solve for the sagitta:

$$s = r - \sqrt{r^2 - y^2} \quad (3)$$

The final equation introduced here, represented in Figure 5, describes the distance from a corneal point along its normal to the reference axis in terms of

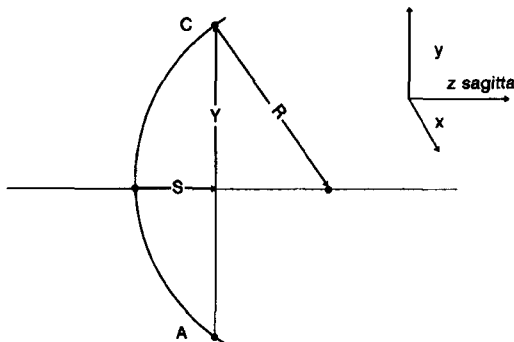


Figure 4. Corneal sagitta (S) for a spherical surface as described by the relation between hemichord (Y) and radius of curvature (R)

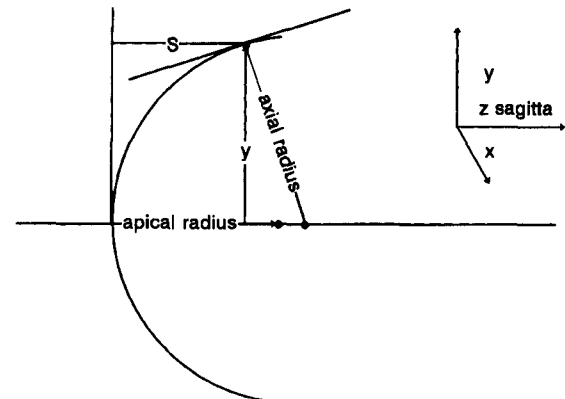


Figure 5. The more general sagittal relation for a non-spherical surface using both the apical (r_o) and axial (r_a) of curvature

the apical radius of curvature (r_o) and the rate of peripheral flattening (p):

$$r_{axial}^2 = r_o^2 + (1-p)y^2 \quad (4)$$

These formulae are commonly encountered in the mathematical derivation of various corneal topography algorithms and corneal models.

FUNDAMENTALS OF CORNEAL TOPOGRAPHY

The keratometer has been the principal instrument for measuring paracentral corneal curvature. Although the keratometer has been invaluable in providing highly repeatable results, it has certain shortcomings which have been exposed by the requirement for quantitative information regarding the very central and mid-peripheral cornea for contact lens and keratorefractive procedures. Videokeratoscopy now challenges keratometry as the clinical standard for measurement of corneal contour. Videokeratoscopes have been found to accurately and repeatedly measure test spheres and human eyes under certain conditions.^{38,80-86} In a single display topographic maps can present radii or power values derived from corneal surface co-ordinates at several thousand data points. Impressive as they may seem,

CT systems are not without limitations and are susceptible to certain types of errors, some due to inherent system assumptions. Instrument hardware and software features such as working distance, faceplate geometry, camera resolution and edge detection limits, as well as the algorithm implemented, determine the instrument sensitivity to focus and alignment error. Instrument defocus and misalignment produce significantly larger errors than those attributed to the various algorithms for regular corneal shapes. There is some debate about the clinical significance of some of the errors involved. Even so, it is well recognised that errors may be introduced into CT outputs due to improper alignment of the instrument.^{10,86} McCarey, Zurawski and O'Shea⁸⁶ reported astigmatism of approximately one dioptre created by 2.5 mm lateral decentration of a test sphere. Less well recognised is the potential error due to any difference between the instrument reference axis and the visual reference axis of the measured eye, that is, relating corneal measurements to visual performance measured at the retinal level using different reference axes.

To appreciate the workings and limitations of videokeratoscope-based corneal

topography systems, it is advantageous to sub-divide the process into three functional steps.

1. image capture
2. computation (reconstruction)
3. display.

Although on first assessment the various commercial units may seem considerably different, all videokeratoscope corneal topography systems work on the same basic optical principles. The video camera captures the image of a Placido disc object formed by the anterior corneal surface acting as a mirror. The digitised video picture is then analysed by imaging software which detects the position of the various rings and determines the distance from a known reference point. This process is repeated for each ring at regular angular intervals, for example, every one or two degrees. The shape of the cornea that produces the video ring images can be calculated by using proprietary algorithms to solve, by iteration, the intractable equations or by using look-up calibration tables. To achieve this numerical result, several significant assumptions regarding eye alignment and relative instrument position are required which impact at different stages in the overall process.

1. Image capture

The image acquisition process is a combination of optical and electronic systems, which captures, by digitising a video signal, the image formed by the corneal surface of a known target, which can then be computer-analysed. This process can be performed using a variety of systems as demonstrated by the different commercial designs.

THE TARGET

Faceplate geometry is best considered in combination with a number of inter-related design factors, most notably, instrument working distance. One of the significant advantages of corneal topography over keratometry is corneal coverage, and this is achieved, in part, due to the size and placement of the ring target. To enable analysis of the entire corneal surface, the target ring must be placed in a position (set by both target size and working distance) so that light from the target is reflected off the cornea towards the videocamera. Analysis of the peripheral cornea is achieved by either a large faceplate or short working distance. Figure 6 illustrates the relationship between working distance and faceplate size for equivalent corneal coverage.

The number of rings, their colour, brightness and spacing are features which may influence instrument performance. Changes to the typical continuous ring design (for example, vertical components, checkerboard patterns) may prove useful in the evaluation of the assumption that the object, the reference axis and image are in the same meridional plane. Various faceplate designs have been tried to minimise the problems encountered focusing on a curved image plane. This and other focusing difficulties (due to the large range of corneal shapes) have been overcome indirectly by the small aperture stop and large depth of focus of the video camera. The resolution of the digitiser in combination with the detector array of the CCD (typically, but not necessarily 512 x 512 pixels) determines how much detail can be captured for analysis. The dichotomy of resolution limits and working distance largely determines the minimum

ring diameter measurable within acceptable accuracy limits by the system.

WORKING DISTANCE

The image capture process is governed by the characteristics of the videocamera, referred to as a charge coupled device (CCD). The effect of positioning error on measured ring image height as a function of working distance is demonstrated in Figures 7a and 7b. It can be seen readily that for the same CCD the potential error is larger for the shorter working distance instrument. Several studies^{27, 87} have demonstrated that longer working distance systems provide greater reproducibility but are more susceptible to shadowing by the subject's eyebrows and nose. Reduced sensitivity to focus and alignment errors has yet to be isolated from the effect of algorithm differences for the various systems used in these studies. Although the shifts are small, a longer working distance moves the Placido ring image further from the entrance pupil plane, which contains the potentially confusing iris detail, improving image contrast and edge detection.

FOCUSING TECHNIQUE

Indirectly related to working distance is the technique used for ensuring adequate

system focus. This not only establishes the appropriate distance between the eye and the videocamera to ensure that the Placido rings are in focus (and at maximum contrast), but positions the instrument so that the mathematical assumption for distance from either ring image plane to camera, or cornea to camera, is satisfied. This cornea to camera distance is vitally important and an error in this distance can translate into significant error in the calculation of the ring image height.

One technique uses two diode laser sources to produce anterior stromal spots which must be precisely overlapped 160 microns within the stroma for the instrument to be perfectly positioned. The addition of a vernier component to this alignment target would make the task less difficult. Another system employs a positioning crosshair within a lighted circle on the temporal limbus, while others rely on triangulation techniques to establish correct focus. Alignment accuracy and reproducibility have been compared to keratometry and between systems, using various reference surfaces and subjects.^{81-86, 88} Operator ease of use has also been considered⁸⁴ enabling description of the likely clinical accuracy of the systems.

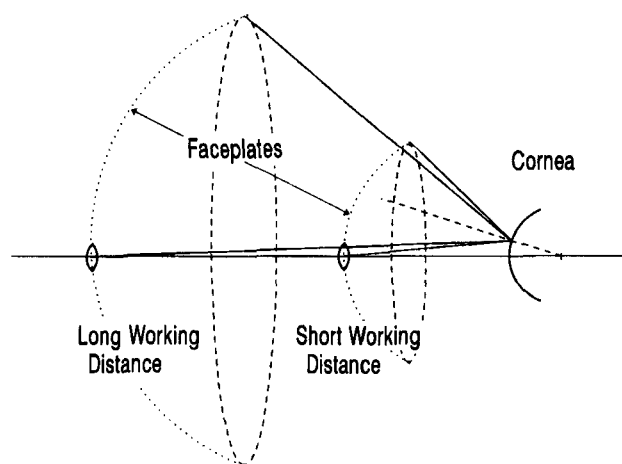


Figure 6. Equivalent corneal coverage achieved by either large faceplate at long working distance or a smaller faceplate with a shorter working distance

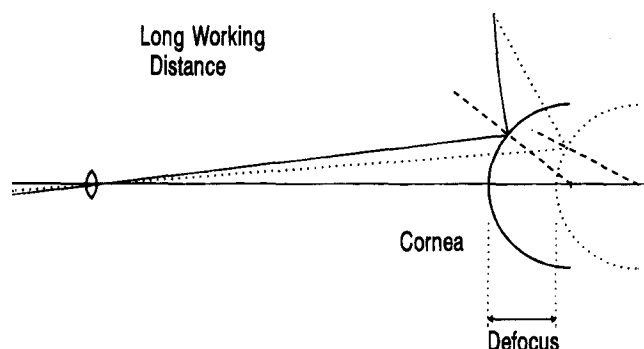


Figure 7a. The interrelation between instrument defocus and image height measurement error for long working distance systems. Not drawn to scale

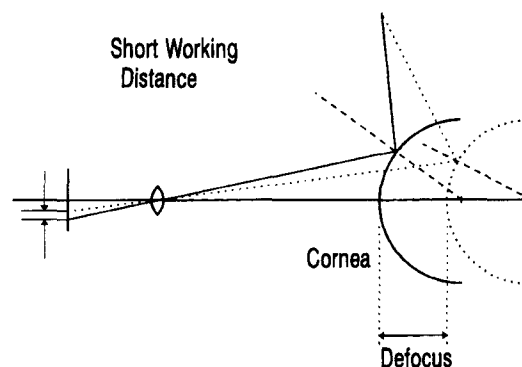


Figure 7b. The interrelation between instrument defocus and image height measurement error for short working distance systems. Not drawn to scale

It is of some benefit to store a record of system focus with each videokeratograph, although some systems now include software to auto-correct small misalignments or report focusing confidence measures.

NUMBER OF RINGS AND CORNEAL COVERAGE

In isolation, the number of rings does not reveal the extent of the corneal area under investigation. The maximum number of useful rings is dependent on CCD and digitiser resolution and it is only when instrument working distance and resolution limits are considered in combination with the shape of the surface being measured, that corneal coverage can be appreciated. Some systems detect the ring edges using both inner and outer ring edges (this more susceptible to errors due to shadows), while others identify the centre of each ring singularly. The current range of commercial videokeratoscope systems measure data points closely surrounding the vertex, which is beyond the capabilities of keratometric techniques. The data for these central readings is noisy compared with those in the mid-periphery for reasons related to percentage errors (ring image size versus pixel size).

Even though limbus to limbus coverage is claimed, palpebral aperture size limits the area of cornea covered by the system. Similarly, eyelashes can interrupt the ring image reducing the extent of peripheral coverage. Figure 8 shows the vignette effect of nose and upper brow clearly seen in the accompanying videokeratoscope image. Short working distance systems are less prone to this effect but are more susceptible to focus and centration errors.

2. Surface reconstruction

IMAGE PROCESSING

Once the image of the Placido rings has been captured by the video camera and digitised, it is analysed using computer imaging software. This process centres around the system's ability to accurately and sensibly identify the position of the ring details. Edge detection algorithms locate the position of the changes in contrast, usually the edge of the thick or the centre of the thin image rings. At times, this process is confounded by discontinuities in iris contrast as the iris image plane is very close to the Placido image plane (Figure 8). Some systems allow manual editing of the ring image and pupil margin to overcome any obviously

incorrect edge detection results, although this feature is not available on all systems.

The ring position is measured from a reference point establishing the instrument axis and it can be appreciated that the sensitivity to error is a function of the system's capabilities in accurately measuring this distance and the reference point.

Dual camera set-ups in some systems are used to increase the resolution limits of the camera-image analysis systems by using a second high magnification camera for the central area. Effectively, this increases the number of pixels, on which inner ring details fall, reducing measurement noise.

The distance from the point of interest on the ring image to the instrument axis reference point (as measured by the number of pixels on the CCD) is then combined with known instrument constants to calculate the corneal co-ordinates and tangent angle. A number of different equations have been published,⁸⁹⁻⁹² but an exact solution is not possible from the given data and several important assumptions are necessary to solve the equations iteratively. Several improvements in the equations have decreased the sensitivity to errors without large increases in computing time.

The general method that CT systems employ is iterative, where, for small segments along each hemi-meridian, different arcs are fitted between the reflection points of each ring to solve the various equations and assumptions. The underlying principle of reflection requires that the angle of incidence equals the angle of reflection for the measured video image height and ring target position.

AREA SAMPLING BIAS

Videokeratoscope systems produce the same number of data points for each ring (for example 360, 256, or 180 per ring), unlike rasterstereographic methods which project a grid onto the corneal surface, setting the number of data points per unit corneal area (although a similar but smaller problem is encountered due to the oblique projection of the grid onto a curved surface). This bias of the number of data points per unit corneal area is important when calculating various surface indices or the best fit spherocylindrical surface from corneal topography data, but is largely negated by the Stiles-Crawford Effect when predicting visual performance measures.

3. Outputs

Presentation of the three-dimensional reconstruction of the corneal surface has progressed from two-dimensional data arrays, through wire mesh models⁹³ to the now familiar colour-coded map. For each of the thousands of data points, either radii or power values can be reported, but some confusion arises because a number of different representations of corneal power are possible. Interpretation of the colour-coded maps must convey to the user either direct information regarding the exact nature of the surface profile or derived information about its optical performance. Roberts¹⁷⁻¹⁹ has pointed out that radii rather than power values might provide a better description of corneal shape, but as explained by Klein and Mandell,²⁴ power units lead to more elegant mathematical expressions and the relationship between visual performance and power units is a well understood general concept. The relative advantages and disadvantages

of each of the various alternatives have been debated and it has been concluded that there are unique practical applications for each representation of power.^{23,24,94}

The different commercial videokeratoscope systems use algorithms which, with few exceptions, produce axial powers as their default power display. It should be noted that at least one system offers what is termed tangential curvature (better described as instantaneous or local curvature) as an extra display. The terms tangential and sagittal have specific optical meanings which should not be confused by the incorrect use of similar terminology in corneal topography because there is an axial and instantaneous power in each of the tangential and sagittal planes. The use of the terms axial and instantaneous powers instead of tangential and sagittal has been favoured²⁴ and allows for the unambiguous treatment of non-radially symmetric surfaces.

The importance of the assumptions regarding a reference axis has been identified in the surface reconstruction stage and in the subsequent calculation of the output maps. Two of the common different representations of corneal power are based on shape properties and provide an approximation of another, namely, refractive power. The three corneal powers have been described²³ as:

1. axial
2. instantaneous
3. refractive.

The two shape-based powers (1 and 2) are calculated using the paraxial approximation and the different measures from the corneal point to a reference point. Axial power is calculated using the distance from the corneal surface point to the system reference axis along the normal to the surface and the surface tangent angle (slope). It is less sensitive to noise than the other forms of corneal power representations. Axial power representations of spherical surfaces show uniform power in the presence of spherical aberration. Instantaneous (local) power is related to curvature and gives the most sensitive measure of local curvature changes in keratoconus. It is independent

of any axis but may provide limited visual performance information as the eye is an axis-based optical system. Refractive power is calculated from ray tracing by using Snell's Law. Although an accurate representation of the refractive characteristics of the anterior corneal surface, further assumptions regarding the crystalline lens, various aberrations and reference axis are necessary. True surface elevation (co-ordinate or x, y, z) has geometric rather than optical significance and is best represented as the deviation from a reference surface, such as in fluorescein pattern modelling.

Scaling

Most systems offer a default scale but with alternative custom and autofit scales. In spite of the claimed benefits of absolute scales and memorising regular colours for quick recognition, these benefits are limited, for example, 1.00 D of cylinder may be clearly visible or completely hidden using a 1.50 D scale. The ideal scale increment varies for different applications and conditions but maximum detail is afforded by increments of about 0.50 D, making best use of the systems' capabilities. Where possible, the same scale should be used consistently.

APPLICATIONS

Computer assisted corneal topography is used in a range of clinical applications ranging from keratorefractive surgery and contact lens fitting, to keratoconus screening and post-operative astigmatism management described below.

Keratorefractive surgery

The region of the cornea that contributes most to foveal image formation overlies the entrance pupil¹¹ and it is this area that is reshaped in keratorefractive surgery. A number of different procedures that alter the refractive power of the eye by creating an optical zone with a new refractive power is available.⁹⁵ The degree of surface smoothness and the surface shape will largely determine the corneal contribution to the final visual outcome.⁹⁶ CT systems are the instrument of choice in

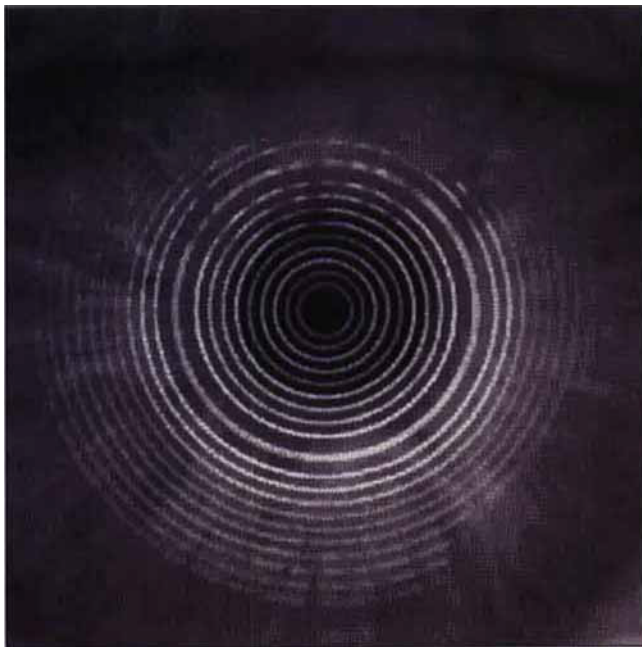


Figure 8. Videokeratograph illustrating ring image and iris detail partly obscured by eyebrows and lids

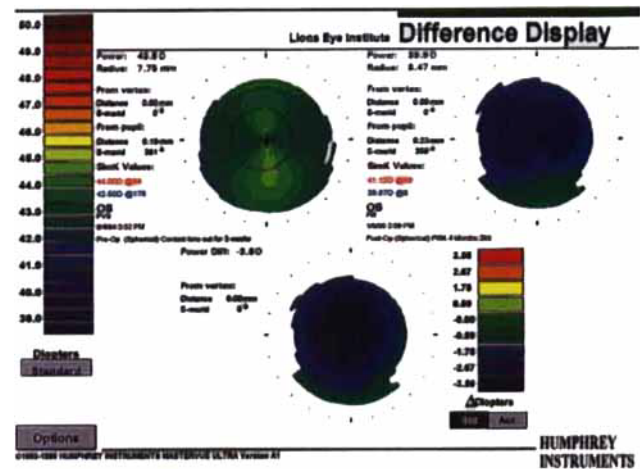


Figure 9. An example of corneal topography power maps pre- and post-operative, and difference plots for excimer photorefractive keratectomy

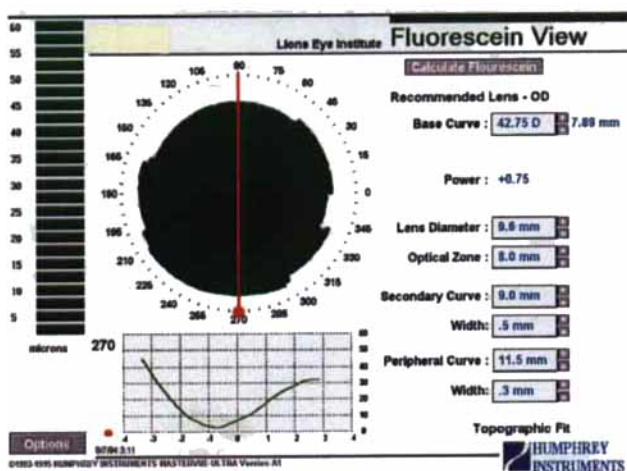


Figure 10. Contact lens fluorescein display simulating tear fluid layer thickness in micrometres

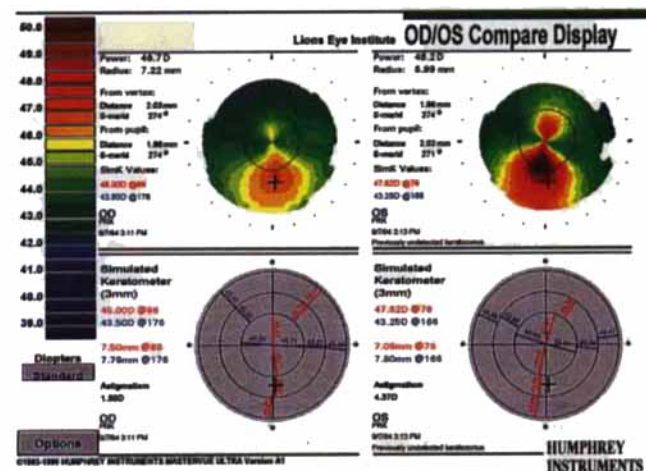


Figure 11. Keratoconic corneal topography power maps for a previously undetected case. Note that condition is more advanced in the left eye

the clinical measurement of corneal contour in the management of keratorefractive procedures.³⁷

The PRK literature^{6-15,57,58,97} has addressed the relationship between visual outcome and centration. Incorrectly centred optical zones can give rise to glare, decreased best corrected visual acuity and decreased contrast sensitivity. Corneal topography power maps, as in Figure 9, are part of the routine pre- and post-operative protocol in keratorefractive surgery, because keratometry is unable to provide sufficient information about the corneal contour.

Contact lens fitting

Many commercial corneal topography systems now offer contact lens software modules which simulate the fluorescein patterns familiar to contact lens practitioners (Figure 10). The selection of rigid contact lens base curves to optimise the fit between the back surface of the lens and the anterior cornea has been based on the concepts of maximum mid-peripheral lens bearing, minimum central clearance and adequate edge lift. Although in their early stages of development, CT empirical systems have performed significantly worse than trial lens fit success rates.⁹⁸ Proprietary in nature, it is likely that most CT systems base their contact lens fitting algorithms on x, y, z data rather than curvature or power values and fail to address the dynamic effects of lens position and flexure or the influence of the eyelids.

This technology is not only used in the selection of lens base curves but plays an important role in the evaluation of contact lens induced corneal topography changes. Warpage, the reversible or permanent changes in corneal topography not associated with oedema, has been observed to persist for months after the cessation of lens wear. Serial videokeratography^{72,73} has been used to monitor such changes in polymethyl-methacrylate, and to a lesser extent, some gas permeable and soft contact lenses.

Keratoconus

The advanced form of keratoconus presents with pathognomonic slit-lamp

signs and a scissored retinoscopic reflex. The subtle changes seen in the early stages or the sub-clinical form of keratoconus typically manifest as asymmetric inferior corneal steepening but are more difficult to categorise. Videokeratography provides the means for detection of such sub-clinical variations in corneal topography (Figure 11) and a number of shape indices have been developed to assist in diagnosis. 'Keratoconus suspect' has been used to describe keratoconus-like topographical patterns in otherwise normal eyes, which typically show inferior steepening.^{67,68,70,99,100} Topographic studies of family members of patients with keratoconus⁶⁹ have lent support to a model with an autosomal dominant mode of inheritance and variable expression. As more data is collected over a longer period, a better understanding will develop of the rate of progression (if any) from the sub-clinical form to the more advanced form of keratoconus. It has been suggested that instantaneous rather than axial power maps better reveal the abrupt curvature changes in disorders like keratoconus.²³ As new software becomes available allowing different output forms, the strength of those assertions will be tested.

Accuracy and reproducibility

As pointed out by Zadnik, Friedman and Mutti,⁸⁸ the corneal topography literature inconsistently reports the reliability and validity of various instruments. They reported the interoccasion repeatability of videokeratoscope measurements and proposed a technique for subdividing the dataset ('the corneal field') for such analyses.

Studies have verified the precision of the various instruments in measuring calibrated spheres and normal corneas under certain conditions.^{81-86,88} The arguments¹⁷⁻¹⁹ regarding the algorithm's spherical bias and inherent error in mapping aspheric surfaces should be considered in the context of the earlier discussion of the relative advantages and disadvantages of the different representations of corneal power. Under certain focus and alignment conditions, the various systems measure to an accuracy of 0.25 D.

CONCLUSION

The measurement of corneal topography has advanced rapidly with improvements in both hardware and software. Videokeratoscopy is a powerful tool which has a range of clinical applications for describing the normal and abnormal corneal shape from evaluating keratorefractive surgery, through fitting contact lenses, to screening for keratoconus. The technology is not without limitations and a thorough understanding of the system design features allows the meaningful clinical interpretation of corneal topography outputs.

REFERENCES

1. Waring GO III. Corneal disorders. Clinical diagnosis and management. In: Leibowitz HM, ed. Corneal Structure and Pathophysiology. Philadelphia: WB Saunders Company, 1984: 3-4.
2. Bennett AG, Rabbetts RB. Clinical Visual Optics. London: Butterworths, 1984: 264-267.
3. Maloney RK, Bogan SJ, Waring GO III. Determination of corneal image-forming properties from corneal topography. *Am J Ophthalmol* 1993; 115: 31-41.
4. Seiler T, Reckmann W, Maloney RK. Effective spherical aberration of the cornea as a quantitative descriptor in corneal topography. *J Cat Refract Surg* 1993; 19 (Suppl): 155-165.
5. Camp JJ, Maguire LJ, Cameron BM, Robb RA. A computer model for the evaluation of the effect of corneal topography on optical performance. *Am J Ophthalmol* 1990; 109: 379-386.
6. Cavanaugh TB, Durrie DS, Riedel SM, Hunkeler JD, Leshner MP. Centration of excimer laser photorefractive keratectomy relative to the pupil. *J Cat Refract Surg* 1993; 19 (Suppl): 144-148.
7. Cavanaugh TB, Durrie DS, Riedel SM, Hunkeler JD, Leshner MP. Topographical analysis of the centration of excimer laser photorefractive keratectomy. *J Cat Refract Surg* 1993; 19 (Suppl): 136-143.
8. Wilson SE, Klyce SD, McDonald MB, Liu JC, Kaufman HE. Changes in corneal topography after excimer laser photorefractive keratectomy for myopia. *Ophthalmology* 1991; 98: 1338-1347.
9. Pande M, Hillman JS. Optical zone centration in keratorefractive surgery. Entrance pupil centre, visual axis, coaxially sighted corneal reflex, or geometric corneal center? *Ophthalmology* 1993; 8: 1230-1237.
10. Hubbe RE, Foulks GN. The effect of poor fixation on computer-assisted topographic

- corneal analysis. Pseudokeratoconus. *Ophthalmology* 1994; 101: 1745-1748.
11. Maloney RK. Corneal topography and optical zone location in photorefractive keratectomy. *J Refract Corneal Surg* 1990; 6: 363-371.
 12. Amano S, Tanaka S, Shimizu K. Topographical evaluation of centration of excimer laser myopic photorefractive keratectomy. *J Cat Refract Surg* 1994; 20: 616-619.
 13. Lin DTC, Sutton HF, Berman M. Corneal topography following excimer photorefractive keratectomy for myopia. *J Cat Refract Surg* 1993; 19 (Suppl): 149-154.
 14. Steinberg EB, Waring GO III. Comparison of two methods of marking the visual axis on the cornea during radial keratotomy. *Am J Ophthalmol* 1983; 96: 605-608.
 15. Klyce SD, Smolek MK. Corneal topography of excimer laser photorefractive keratectomy. *J Cat Refract Surg* 1993; 19 (Suppl): 122-130.
 16. Fincham WHA, Freeman MH. Optics. 10th ed. London; Butterworths, 1990.
 17. Roberts C. Analysis of the inherent error of the TMS-1 Topographic Modeling System in mapping a radially aspheric surface. *Cornea* 1995; 14: 258-265.
 18. Roberts C. Characterization of the inherent error in a spherically-biased corneal topography system in mapping a radially aspheric surface. *J Refract Corneal Surg* 1994; 10: 103-111.
 19. Roberts C. The accuracy of 'power' maps to display curvature data in corneal topography systems. *Invest Ophthalmol Vis Sci* 1994; 35: 3525-3532.
 20. Mandell RB. Comments on Dr Roberts' article entitled 'Characterization of the inherent error in a spherically-biased corneal topography system in mapping a radially aspheric surface'. *J Refract Corneal Surg* 1994; 10: 112.
 21. Applegate RA. Comments on Dr Roberts' article entitled 'Characterization of the inherent error in a spherically-biased corneal topography system in mapping a radially aspheric surface'. *J Refract Corneal Surg* 1994; 10: 113-114.
 22. Cohen KL, Tripoli NK, Holmgren DE, Coggins JM. Assessment of the power and height of radial aspheres reported by a computer-assisted keratoscope. *Am J Ophthalmol* 1995; 119: 723-732.
 23. Klein SA, Mandell RB. Shape and refractive powers in corneal topography. *Invest Ophthalmol Vis Sci* 1995; 36: 2096-2109.
 24. Klein SA, Mandell RB. Axial and instantaneous power conversion in corneal topography. *Invest Ophthalmol Vis Sci* 1995; 36: 2155-2159.
 25. Uozato H, Guyton DL. Centering corneal surgical procedures. *Am J Ophthalmol* 1987; 103: 264-275.
 26. Mandell RB. Locating the corneal sighting center from videokeratography. *J Refract Surg* 1995; 11: 253-259.
 27. Mandell RB. The enigma of the corneal contour. *CLAO J* 1992; 18: 267-273.
 28. Campbell MCW, Harrison EM, Simonet P. Psychophysical measurement of the blur on the retina due to optical aberrations of the eye. *Vision Res* 1990; 30: 1587-1602.
 29. Thibos LN, Bradley A, Still DL, Zhang X, Howarth PA. Theory and measurement of ocular chromatic aberration. *Vision Res* 1990; 30: 33-49.
 30. Arffa RC, Warnicki JW, Rehkopf PG. Corneal topography using rasterstereography. *J Refract Corneal Surg* 1989; 5: 414-417.
 31. Joseph TK, Kartha CP. The Kerala decentration meter. A new method and devise for fitting the optical of spectacle lenses in the visual axis. *Acta Ophthalmol* 1982; 60 (Suppl 151): 5-30.
 32. Mandell RB. Apparent pupil displacement in videokeratography. *CLAO J* 1994; 20: 123-127.
 33. Ludlam WM. Photographic-ophthalmometric and clinical investigations of corneal refraction. *Am J Optom* 1966; 43: 143-214.
 34. Clark BAJ. Keratometry: A Review. *Aust J Optom* 1973; 56: 94-100.
 35. Mandell RB. Methods to measure the peripheral corneal curvature, part 3: Ophthalmometry. *J Am Optom Assoc* 1963; 33: 889-892.
 36. Mandell RB. Reflection point ophthalmometry. A method to measure corneal contour. *Am J Optom* 1962; 39: 513-537.
 37. Klyce SD, Wilson SE. Methods of analysis of corneal topography. *J Refract Corneal Surg* 1989; 5: 368-371.
 38. Wilson SE, Klyce SD. Advances in the analysis of corneal topography. *Surv Ophthalmol* 1991; 35: 269-277.
 39. Sanders DR, Gills JP, Martin RG. When keratometric measurements do not accurately reflect corneal topography. *J Cat Refract Surg* 1993; 19 (Suppl): 131-135.
 40. Belin MW. Intraoperative raster photogrammetry—the PAR Corneal Topography System. *J Cat Refract Surg* 1993; 19 (Suppl): 188-192.
 41. Clark BAJ. Systems for describing corneal topography. *Aust J Optom* 1973; 56: 48-56.
 42. Clark BAJ. Variations in corneal topography. *Aust J Optom* 1973; 56: 399-413.
 43. Mandell RB, St Helen R. Position and curvature of the corneal apex. *Am J Optom Arch Am Acad Optom* 1969; 46: 25-29.
 44. Lopping B, Weale RA. Changes in corneal curvature following ocular convergence. *Vision Res* 1965; 5: 207-215.
 45. Fairmaid JA. The constancy of corneal curvature. An examination of corneal response to changes in accommodation and convergence. *Br J Physiol Opt* 1959; 16: 2-23.
 46. Bannon RE. Near binocular problems—astigmatism and cylcophoria. *Ophthalmol Opt* 1971; 11: 158-168.
 47. Brzezinski MA. Review: Astigmatic accommodation (sectional accommodation)—a form of dynamic astigmatism. *Aust J Optom* 1982; 65: 5-11.
 48. Kame RT, Jue TS, Shigekuni DM. A longitudinal study of corneal astigmatism changes in Asian eyes. *J Am Optom Assoc* 1993; 64: 215-219.
 49. Phillips RA. Changes in corneal astigmatism. *Am J Optom Arch Am Acad Optom* 1952; 29: 379-380.
 50. Lyle WM. Changes in corneal astigmatism with age. *Am J Optom Arch Am Acad Optom* 1971; 48: 467-478.
 51. Fledelius HC, Stubgaard M. Changes in refraction and corneal curvature during growth and adult life. *Acta Ophthalmol* 1986; 64: 487-491.
 52. Miller D. Pressure of the lid on the eye. *Arch Ophthalmol* 1967; 78: 328-330.
 53. Grey C, Yap M. Influence of lid position on astigmatism. *Am J Optom Physiol Opt* 1986; 63: 966-969.
 54. Mandell RB, St. Helen R. Stability of the corneal contour. *Am J Optom Arch Am Acad Optom* 1968; 45: 797-806.
 55. Carney LG, Clark BAJ. Experimental deformation of the *in vivo* cornea. *Am J Optom* 1971; 48: 28-34.
 56. Bowman KJ, Smith G, Carney LG. Corneal topography and monocular diplopia following near work. *Am J Optom Physiol Opt* 1978; 55: 818-823.
 57. Brancato R, Carones F, Venturi E, Bertuzzi A. Corticosteroids vs Diclofenac in the treatment of delayed regression after myopic photorefractive keratectomy. *J Refract Corneal Surg* 1993; 9: 376-378.
 58. Cantera E, Cantera I, Olivieri L. Qualitative evaluation of photorefractive keratectomy with computer assisted corneal topography. *J Refract Corneal Surg* 1994; 10 (Suppl): 296-298.
 59. Patel S, Marshall J, Fitzke FW, Gartry DS. The shape of the corneal apical zone after excimer photorefractive keratectomy. *Acta Ophthalmol* 1994; 72: 588-596.
 60. Reidy JJ, McDonald MB, Klyce SD. The corneal topography of epikeratophakia. *J Refract Corneal Surg* 1990; 6: 26-31.
 61. Terrell J, Bechara SJ, Nesburn A, Waring GO III, Macy J, Maloney RK. The effect of globe fixation on ablation zone centration in photorefractive keratectomy. *Am J Ophthalmol* 1995; 119: 612-619.
 62. Hayashi K, Nakao F, Hayashi F. Topographic analysis of early changes in corneal astigmatism after cataract surgery. *J Cat Refract Surg* 1993; 19: 43-47.
 63. Hayashi K, Nakao F, Hayashi F. Corneal topographic analysis of superolateral incision cataract surgery. *J Cat Refract Surg* 1994; 20: z392-399.
 64. Rosen WJ, Mannis MJ, Brandt JD. The effect of trabeculectomy on corneal topogra-

- phy. *Ophthalmic Surg* 1992; 23: 395-398.
65. Walland MJ, Stevens JD, Steele ADM. The effect of recurrent pterygium on corneal topography. *Cornea* 1994; 13: 463-464.
66. Maguire LJ, Bourne WM. Corneal topography of early keratoconus. *Am J Ophthalmol* 1989; 108: 107-112.
67. Maeda N, Klyce SD, Smolek MK, Thompson HW. Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci* 1994; 35: 2749-2757.
68. Maguire LJ, Lowry JC. Identifying progression of subclinical keratoconus by serial topography analysis. *Am J Ophthalmol* 1991; 112: 41-45.
69. Rabinowitz YS, Garbus J, McDonnell PJ. Computer-assisted corneal topography in family members of patients with keratoconus. *Arch Ophthalmol* 1990; 108: 365-371.
70. Wilson SE, Lin DTC, Klyce SD. Corneal topography of keratoconus. *Cornea* 1991; 10: 2-8.
71. Levin S, Carson CA, Garrett SK, Taylor HR. Prevalence of central islands after excimer laser refractive surgery. *J Cat Refract Surg* 1995; 21: 21-26.
72. Wilson SE, Lin DTC, Klyce SD, Reidy JJ, Insler MS. Topographic changes in contact lens-induced corneal warpage. *Ophthalmology* 1990; 97: 734-744.
73. Maeda N, Klyce SD, Hamano H. Alteration of corneal asphericity in rigid gas permeable contact lens induced warpage. *CLAO J* 1994; 20: 27-31.
74. Levenson DS. Changes in corneal curvature with long-term PMMA contact lens wear. *CLAO J* 1983; 9: 121-125.
75. Mandell RB, St Helen R. Mathematical model of the corneal contour. *Br J Physiol Opt* 1971; 26: 183-197.
76. Burek H, Douthwaite WA. Mathematical models of the general corneal surface. *Ophthalm Physiol Opt* 1993; 13: 68-72.
77. Baker TY. Ray tracing through non-spherical surfaces. *Proc Phys Soc*. 1943; 55: 361.
78. Guillon M, Lydon DPM, Wilson C. Corneal topography: a clinical model. *Ophthalm Physiol Opt* 1986; 6: 47-56.
79. Martin DK, Holden BA. A new method for measuring the diameter of the *in vivo* human cornea. *Am J Optom Physiol Opt* 1982; 59: 436-441.
80. Hannush SB, Crawford L, Waring GO III. Accuracy and precision of keratometry, photokeratoscopy, and corneal modelling on calibrated steel balls. *Arch Ophthalmol* 1989; 107: 1235-1239.
81. Tsilimbaris MK, Vlachonikolis IG, Siganos DD, Makridakis G, Pallikaris IG. Comparison of keratometric readings as obtained by Javal Ophthalmometer and corneal analysis system (EyeSys). *J Refract Corneal Surg* 1991; 7: 368-373.
82. Legeais J-M, Ren Q, Simon G, Parel J-M. Computer-assisted corneal topography: Accuracy and reproducibility of the topographic modeling system. *J Refract Corneal Surg* 1993; 9: 347-357.
83. Maguire LJ, Wilson SE, Camp JJ, Verity S. Evaluating the reproducibility of topography systems on spherical surfaces. *Arch Ophthalmol* 1993; 111: 259-262.
84. Antalis JJ, Lembach RG, Carney LG. A comparison of the TMS-1 and the Corneal Analysis System for the evaluation of abnormal corneas. *CLAO J* 1993; 19: 58-63.
85. McCarey BE, Amos CF. Topographic evaluation of toric soft contact lens correction. *CLAO J* 1994; 20: 261-265.
86. McCarey BE, Zurawski CA, O'Shea DS. Practical aspects of a corneal topography system. *CLAO J* 1992; 18: 248-254.
87. Nieves JE, Applegate RA. Alignment errors and working distance directly influence the accuracy of corneal topography measurements. *ARVO Abstracts. Invest Ophthalmol Vis Sci* 1992; 33 (Suppl): 993.
88. Zadnik K, Friedman NE, Mutti DO. Repeatability of corneal topography: The 'corneal field'. *J Refract Surg* 1995; 11: 119-125.
89. Wang J, Rice DA, Klyce SD. A new reconstruction algorithm for improvement of corneal topography analysis. *J Refract Corneal Surg* 1989; 5: 370-387.
90. Klein SA. A corneal topography algorithm that produces continuous curvature. *Optom Vis Sci* 1992; 69: 829-834.
91. Doss JD, Hutson RL, Rowsey JJ, Brown DR. Method for calculation of corneal profile and power distribution. *Arch Ophthalmol* 1981; 99: 1261-1265.
92. van Saarloos PP, Constable IJ. Improved method for calculation of corneal topography for any photokeratoscope geometry. *Optom Vis Sci* 1991; 68: 960-965.
93. Klyce SD. Computer-assisted corneal topography. High-resolution graphic presentation and analysis of keratoscopy. *Invest Ophthalmol Vis Sci* 1984; 25: 1426-1435.
94. Salmon TO, Horner DG. Comparison of elevation, curvature, and power descriptors for corneal topographic mapping. *Optom Vis Sci*. 1995; 72: 800-808.
95. Munnerlyn CR, Koons SJ, Marshall J. Photorefractive keratectomy: A technique for laser refractive surgery. *J Cat Refract Surg* 1988; 14: 46-52.
96. Wilson SE, Klyce SD. Quantitative descriptors of corneal topography. *Arch Ophthalmol* 1991; 109: 349-353.
97. Keller PR, van Saarloos PP, Yellachich D. A computer simulation of centration effects on cornea topography analysis of excimer PRK ablations. *Cornea* 1997; 16: 54-63.
98. Kennedy DW, Mintle LR, Schneider CM. The computer vs the optometrist: RGP diagnostic trial fitting procedures compared to the EyeSys computer videokeratography system. *Abstracts. Optom Vis Sci* 1994; 71 (Suppl): 92.
99. Bowman CB, Thompson KP, Stulting RD. Refractive keratotomy in keratoconus suspects. *J Refract Surg* 1995; 11: 202-206.
100. Nesburn AB, Bahri S, Salz J, Rabinowitz YS, Maguen E, Hofbauer J, Berlin M, Macy JJ. Keratoconus detected by videokeratography in candidates for photorefractive keratectomy. *J Refract Surg* 1995; 11: 194-201.

Author's address:

Peter Keller
Lions Eye Institute
2 Verdun Street
Nedlands WA 6009
AUSTRALIA